

How Accurate is 3DLOOK?

A Deep Dive into AI-Powered Body
Measurement Precision

February 2025

Dataset Overview: Structure, Benchmarking, and Considerations

A real-world study utilizing multiple customer scan events, benchmarked against expert manual measurements to assess consistency and precision

Dataset Structure and Methodology:

 **Subjects:** All individuals scanned under controlled conditions.

 **Dataset Composition:** The dataset includes multiple real-world datasets, with each person undergoing five repeated scans for repeatability analysis.

 **Measurement Types:** Categorized into front linear, side linear, and volume-based parameters.

 **Comparison Benchmark:** AI-generated measurements were compared against multiple sets of manual measurements taken by expert pattern makers.
Note: Different people may measure the same body part differently, introducing variability.

Benchmarking Against Manual Measurements: Considerations & Limitations

Challenge	Manual Measurement Issues	How AI Improves It
Human Variability	Subjective, hand placement errors	AI eliminates manual inconsistency
Soft Tissue Areas	Fluctuations in breathing & posture	AI standardizes measurements
Repeatability	Same person gets different results	AI significantly improves repeatability but minor scan-to-scan variations exist

Why This Dataset Matters:

 **Reflects Real-World Accuracy:** Real-world customer scans ensure results are reflective of everyday use and variability, offering a more practical measure of performance.

 **Identifies Variability:** Multiple scans per person highlight natural measurement differences for AI refinement.

 **Validates AI Performance:** Comparisons to manual measurements reveal strengths and areas for improvement.

This dataset serves as a robust foundation for evaluating and enhancing 3DLOOK’s measurement precision.

Measurement Precision and Error Analysis

Absolute Difference Analysis: Measuring AI Accuracy Against Manual Data

Evaluating AI-based body measurements by calculating absolute error margins between predicted and manual values.

What is Absolute Difference?

- ✓ Measures numerical difference between AI and manual measurements.
- ✓ Expressed in cm for direct comparison instead of statistical variance.

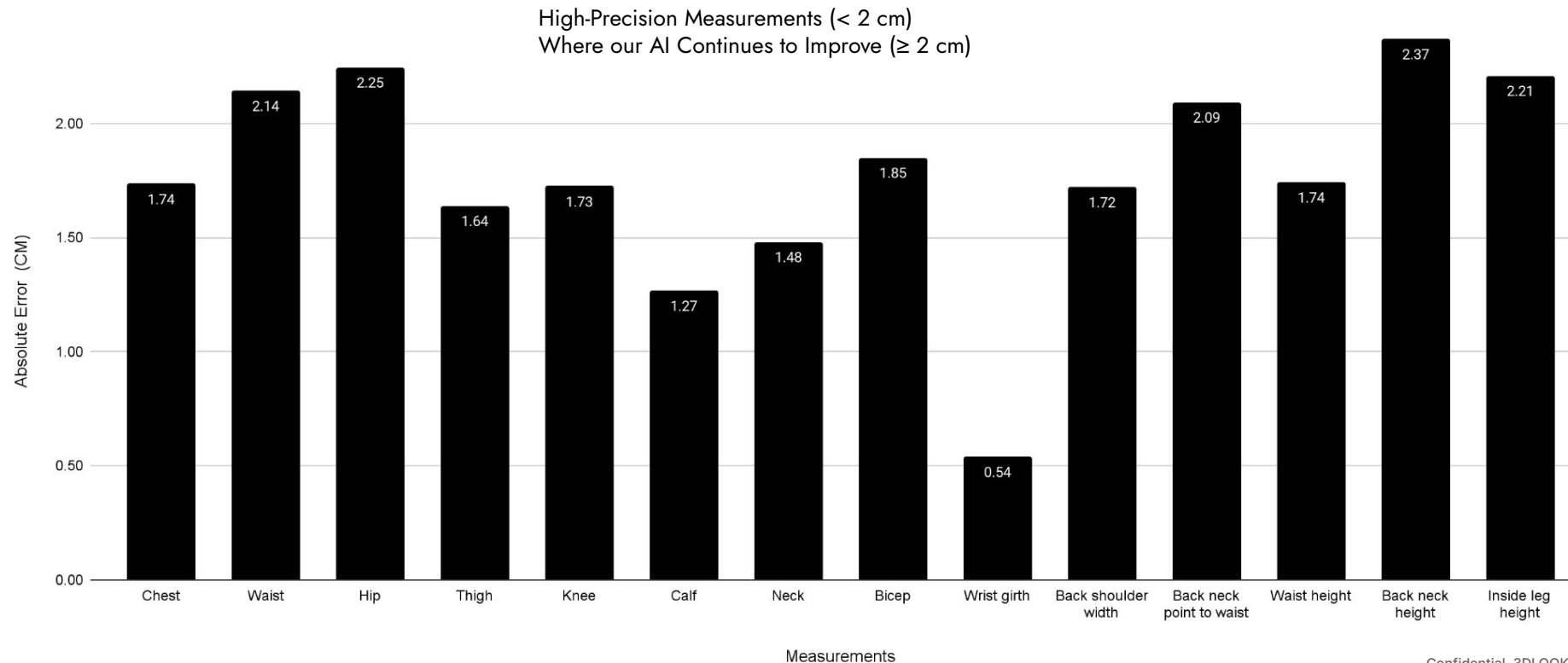
Why Use This Instead of Statistical Variance?

- ✓ **More intuitive:** Real-world units make error margins easier to interpret than variance percentages.
- ✓ **Directly quantifies precision:** Smaller difference = Higher accuracy.
- ✓ **Better for AI improvement tracking:** Identifies specific measurement deviations for model refinement.

Absolute Difference Analysis confirms AI-driven body measurement improves precision and reduces human error, making it more reliable for real-world applications.

Breaking Down AI Measurement Accuracy: Absolute Error by Body Part

Comparison of 3DLOOK Measurements to Manual Measurements: The Smaller the Difference, the Higher the Accuracy.



Overall Accuracy Summary

Conclusion Based on Absolute Difference

Analysis: Reliable & Continuously Advancing

AI measurements achieve high accuracy in key body areas, aligning with top industry standards for mobile body scanning (1.5–2.2 cm error range).

✓ **Highest accuracy:** Wrist girth (0.54 cm).

✓ **Strong precision in key metrics:** Calf (1.27 cm), Neck (1.48 cm), Thigh (1.64 cm), Knee (1.73 cm).

✓ **Critical Areas with Strong Performance & Ongoing Refinement:** Waist (1.74 cm), Hip (2.25 cm) – Already within top industry accuracy standards, with continued improvements in AI pose detection, AI clothing detectors, and soft tissue modeling

AI body scanning delivers consistent, reliable measurements for fitness, telehealth, and apparel applications. Ongoing refinements will enhance accuracy in height-dependent metrics.

Repeatability Analysis: Measuring Stability & Consistency in AI-Generated Measurements

How We Measure AI Consistency: Repeatability Analysis Methodology

AI body measurements were analyzed using variance, a statistical measure of consistency. Lower variance means higher repeatability—ensuring AI delivers stable, reliable body metrics.

✓ **Step 1:** Collected body measurement data across different body parts, categorized into front, side, and volume-based metrics.

✓ **Step 2:** Calculated variance to determine how consistently the AI replicates body measurements.

✓ **Step 3: Classified results into three accuracy levels based on variance:**

✓ **Low Variance (High Stability)** → Highly repeatable, reliable AI measurements

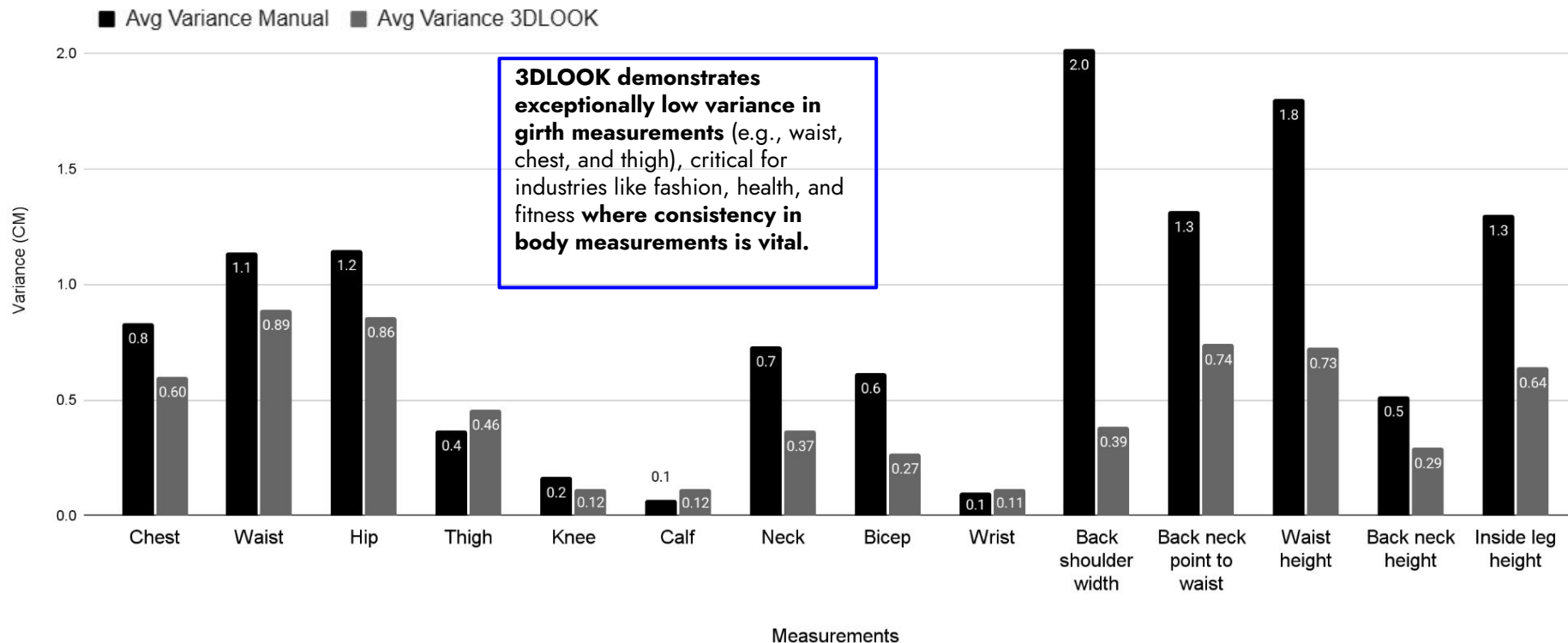
⚠ **Moderate Variance (Acceptable Accuracy)** → Stable with minor fluctuations

✗ **High Variance (Inconsistency)** → Areas needing further refinement

✓ **Step 4:** Identified patterns and drew conclusions→ Which measurements are most and least stable?

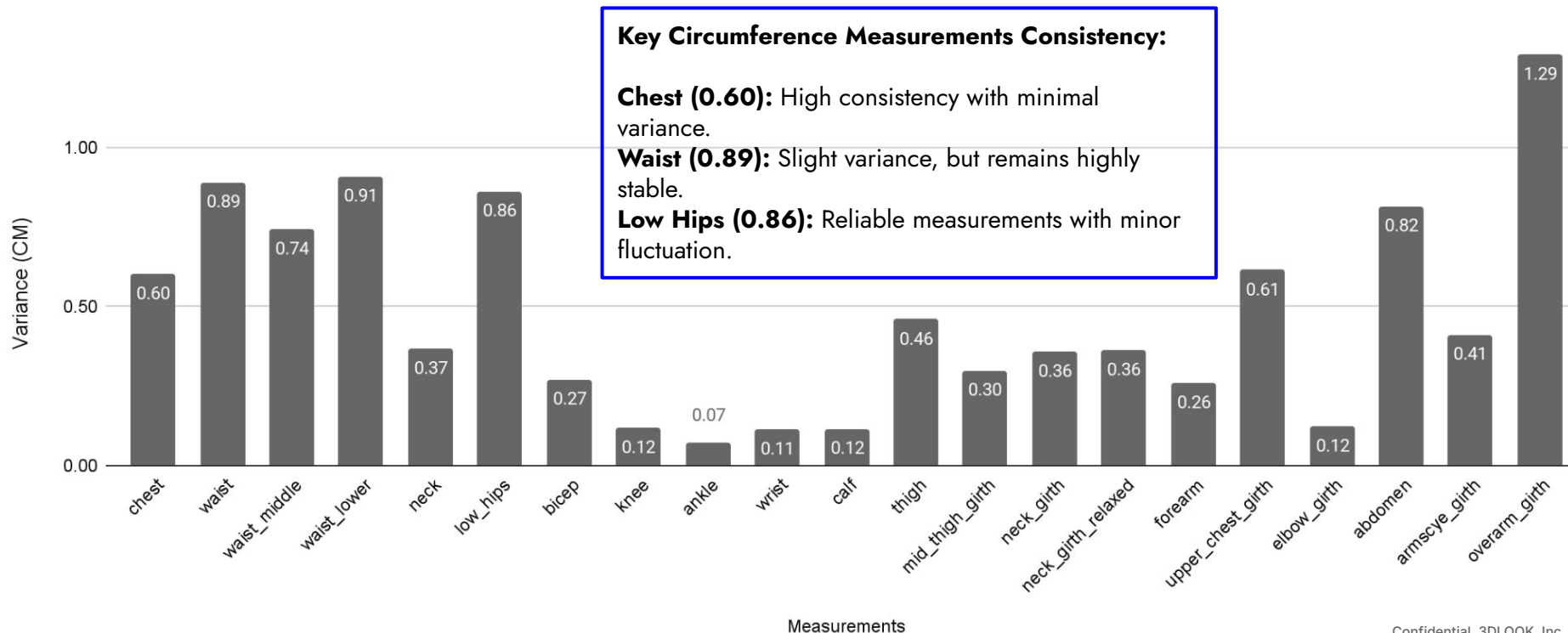
AI vs. Manual Measurements: Consistency and Stability in Body Metrics

3DLOOK delivers consistent, reliable measurements with significantly lower variability across key body metrics, based on real world customer scanning data.



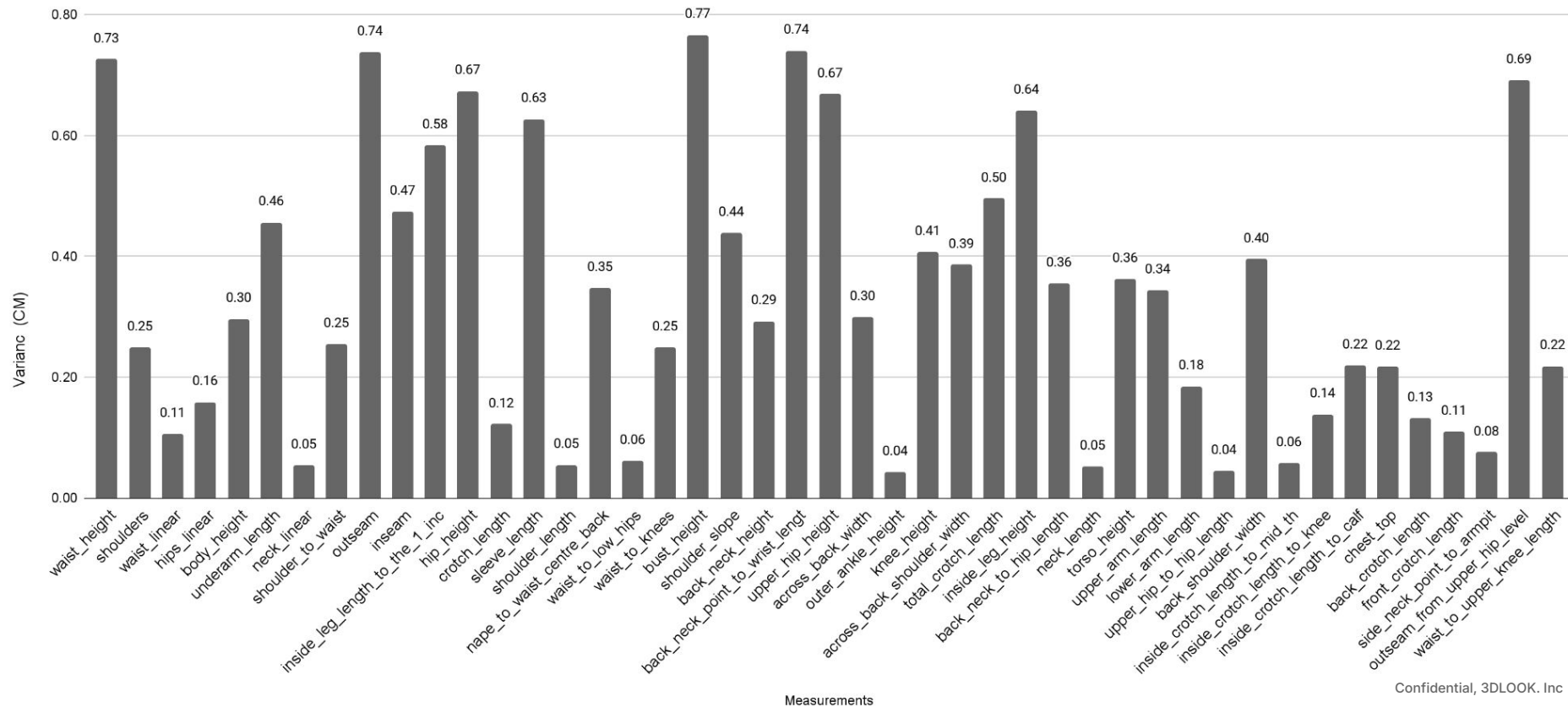
Stability and Repeatability Across Body Metrics: Circumference

3DLOOK provides highly stable and repeatable circumference measurements, ensuring precise and reliable scanning for body girth across various dimensions, with minimal variability even across multiple scans.



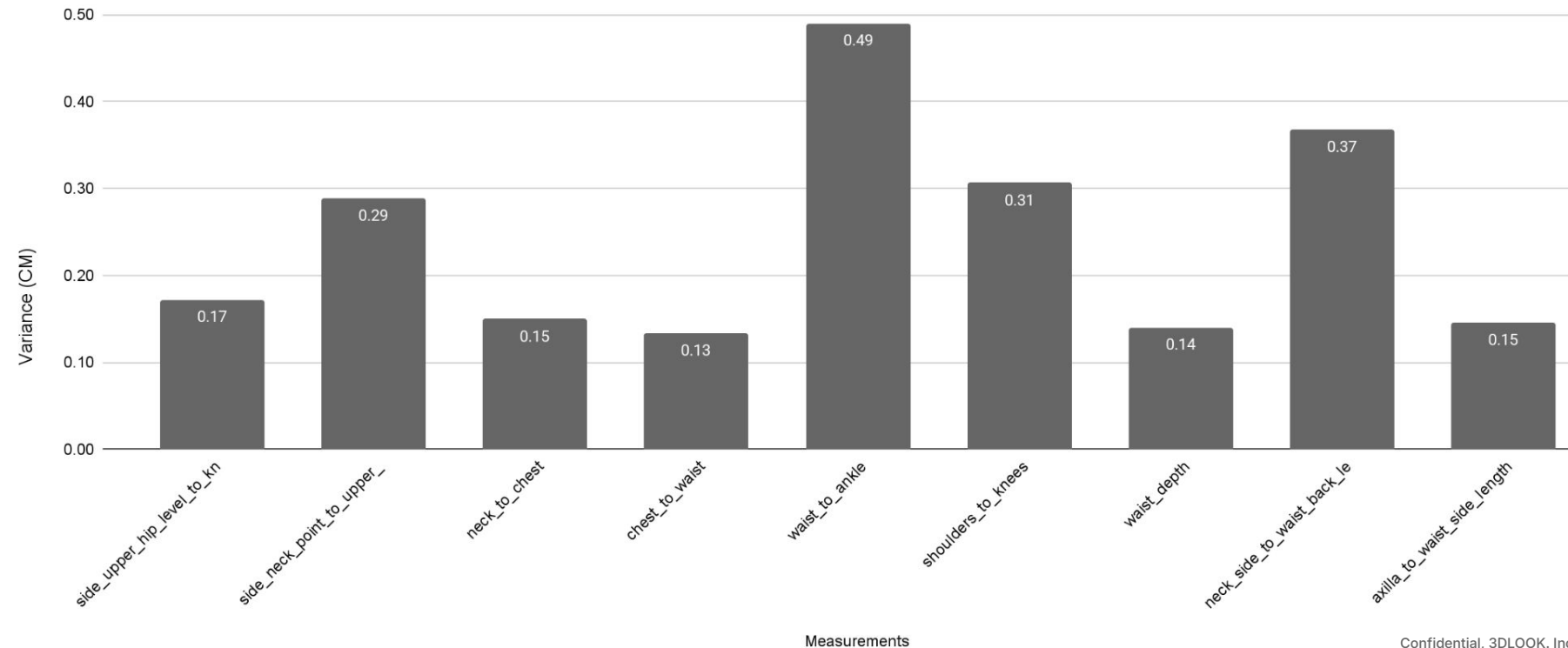
Stability and Repeatability Across Body Metrics: Front Linear

3DLOOK ensures high repeatability in front linear body measurements with minimal variance across most metrics and slight fluctuations due to natural posture variations.



Stability and Repeatability Across Body Metrics: Side Linear

3DLOOK ensures high repeatability in torso and upper body measurements, with minor variability in leg and spine measurements due to natural posture shifts.



Overall Repeatability Summary

Overall Summary

1. **Exceptional accuracy of key anatomical points:** Joint-based and clearly defined anatomical points (knees, ankles, wrists, shoulders) demonstrate high stability and precision, ensuring reliable and consistent results.
2. **Maximum adaptability of soft tissue measurements:** Girth-based and soft tissue measurements effectively account for individual body characteristics, delivering a highly personalized and precise approach for each user.
3. **High consistency in long vertical measurements:** Long vertical measurements demonstrate strong consistency, with ongoing algorithm enhancements aimed at further improving accuracy.
4. **Precise torso and mid-body measurements:** Torso and mid-body measurements show stable and high-quality results, with ongoing refinements to further ensure increased accuracy.

Factors Influencing Measurement Variability

Factors Influencing Measurement Variability

1. Optimized Pose and Image Quality Factors

- **Consistent posture for precise measurements:** Maintaining a stable body position enhances accuracy, with even small adjustments (e.g., arm placement) being effectively managed by ongoing optimizations.
- **Smart perspective adaptation:** Smartphone cameras use wide-angle lenses, and continuous improvements in image processing ensure precise measurement adjustments, especially in key areas like the underbust, waist, and hips.
- **Enhanced lighting and background adaptability:** Advanced algorithms effectively handle various lighting conditions and backgrounds, with further refinements boosting image analysis accuracy.

2. Continuous Algorithm Advancements for Superior Data Processing

- **AI-Powered Pose Detection:** Live visual guidance, voice, and text prompts ensure correct alignment for accurate scans without retakes.
- **AI-Powered Clothing Detector:** AI-powered model assesses clothing tightness and adjusts the 3D output for more accurate measurements.
- **Enhanced model calibration:** Continuous optimization of algorithmic parameters further sharpens measurement precision.
- **Seamless adaptation to diverse body shapes:** Ongoing model refinements ensure even greater accuracy for individuals with unique body proportions.

Conclusion

Measurement consistency continues to improve with advancements in algorithm accuracy, enhanced user guidance, refined perspective correction techniques, and AI-powered clothing detection. These innovations continue to drive higher precision, ensuring reliable and seamless body measurement and composition solutions.

Conclusion: Proven
Accuracy and Consistency
in Real-World Applications

Conclusion

3DLOOK's AI-powered body measurement technology demonstrates approximately **96-97% accuracy** when compared to manual measurements across all body metrics, with a typical error margin of 1.5–2.0 cm.

For repeatability, the technology shows over **95% consistency, with a variance of less than 1 cm** across repeated scans, ensuring high stability and reliability across the full range of body measurements.

This data comes from a real-world study utilizing multiple customer scan events, benchmarked against expert manual measurements **to assess consistency and precision, ensuring that results are reflective of actual use cases** rather than controlled lab conditions.

Ongoing advancements in AI algorithms continue **to improve accuracy and reduce variability** in real-world applications.